

Postdoctoral Position (18 months)

Machine Learning for d- and f-Metal Catalysis

Project MAD-CAT

MAchine learning-based strategies for d- and f-metals CATalyst optimization and discovery: a path toward sustainable (de)polymerization

Location: LPCNO (Laboratoire de Physique et Chimie des Nano-Objets, UMR 5215), Université de Toulouse, France

Team: MPC — Modélisation Physique et Chimique

Duration: 18 months, fully funded (NanoX program)

Supervision: Profs. Laurent Maron & Romuald Poteau

Start date: As soon as possible / negotiable

Context

Chemical catalysis underpins industrial chemistry, energy conversion, environmental remediation and biological processes. Yet it remains a complex, multiscale phenomenon in which electronic structure, molecular dynamics and thermodynamics are tightly intertwined. Density Functional Theory (DFT) has become the dominant framework for studying catalytic systems at a feasible cost, but the exhaustive exploration of reaction pathways remains a daunting task: most studies are restricted to a handful of emblematic mechanisms and rely on the manual, tedious identification of transition states. This "trial-and-error" bottleneck limits the discovery of unexpected pathways and prevents a truly systematic screening of chemical space.

The **MPC team** has built a deep-rooted expertise in the DFT-based study of reaction mechanisms and of the geometric and electronic structure of *d*- and *f*-block metal complexes, most often in close collaboration with experimental groups. This work is backed by a strong track record of roughly 600 peer-reviewed publications. Crucially, decades of computational effort have accumulated into a unique "gold mine" of structural and energetic data, *i.e.* hundreds of transition states and reaction intermediates, that has, so far, remained under-exploited as a machine-learning resource.^[1,2]

MAD-CAT will convert this legacy into a machine-learning-ready, curated and open-access dataset, and use it to bridge traditional quantum chemistry with state-of-the-art machine learning. The project builds on the integration of Machine Learning, Data Science and Explainable AI (xAI) within the MPC team, materialised in the [pyPhysChem open-source platform](#) and in the ongoing co-supervision of several PhD projects at the interface of chemistry and data science.^[3] The recent, successful DFT/ML study of styrene polymerization by group-III metallocenes^[3c] provides a validated proof-of-concept for the interpretable, low-data workflow that MAD-CAT will now generalise.

Recent years have seen an explosion of ML applications for catalytic discovery, but most efforts target relatively simple organic molecules or idealised surfaces.^[4] A significant gap remains for the complex coordination chemistry of *d*- and *f*-block metals, with their variable oxidation states, high coordination numbers and subtle electronic effects. MAD-CAT is positioned squarely in this gap, and its distinctiveness lies in the systematic exploitation of a database unrivalled in its diversity of heavy-metal complexes and in its direct relevance to real-world catalytic applications.

Scientific Objectives

The recruited researcher will help deploy a **three-pillar strategy** that leverages high-quality, quantum-chemistry-based data to navigate the chemical space of heavy-metal complexes with high accuracy:

1. **Targeted surrogate models (low-data regime).** Development of reaction-specific surrogate models based on a form of "chemical essentialization": instead of an atomistic description, coarse-grained yet physically meaningful descriptors (e.g. global steric bulk, Lewis acidity of the metal centre) capture the core drivers of reactivity, enabling high-precision energy predictions where data is sparse and avoiding overfitting. Interpretability is ensured through xAI (SHAP analysis).
2. **Generative latent-space discovery.** Design of ligands and catalysts using a Variational Autoencoder (VAE) coupled to an Artificial Neural Network (ANN). The VAE encodes and decodes the chemical space (SMILES or graph-based representations of the in-house database) while the ANN maps the latent space to relative energies, enabling the "inverse design" of catalysts with target transition-state energies before rigorous computational validation.
3. **Equivariant ML potentials (the core of MAD-CAT).** Fine-tuning of a MACE equivariant interatomic potential on the extracted geometries, supplemented by an active-learning loop with committee-based uncertainty and on-the-fly DFT refinement. Once trained, the potential is coupled with Maeda's AFIR method and automatic TS-search techniques (e.g. Nudged Elastic Band) to automate the discovery of entire reaction networks, bypassing the manual search for transition states.

These tools will be deployed on two application axes where the MPC team holds recognised expertise:

- **CO₂ reduction** — acceleration of the search for catalysts able to lower the high kinetic barriers of CO₂ activation on transition metals, lanthanides and actinides. This axis will also serve as a stringent benchmark and validation field for the fine-tuned MACE potential, and as a testbed for mapping exhaustive reaction networks (competitive side-reactions, off-cycle resting states) previously out of reach with standard DFT.
- **Stereoselective polymerization** — extension of the surrogate workflow beyond the first monomer insertion to second and third insertions, accounting for tacticity control. Predictive modelling of insertion sequences (energetics and orientation) will provide insight into stereocontrol and the design of catalysts tailored for syndiotactic, isotactic or atactic polymers.

As a **medium-to-long-term goal**, the workflows, databases and ML potentials developed during the project will lay the foundations for tackling the **catalytic depolymerization of plastics** (polyesters, polyolefins), a major challenge for the circular economy and an ideal playground for large-scale ML potentials working at near-quantum accuracy.

Deliverables

- **A curated database** of *d*- and *f*-element complexes, reaction intermediates and transition states, consistently optimised at the DFT level, with data augmentation (perturbed geometries for non-zero forces) for MACE training.
- **Fine-tuned MACE equivariant potentials** released together with a detailed description of the chemical space covered (metals and oxidation states, ligands, energies, forces).
- **Auxiliary Surrogate models** for specific reactions and catalysts^[3c]

Candidate Profile

- PhD in computational/theoretical chemistry, computational materials science or a closely related field.

- **Priority expertise:** MACE (or other equivariant / ML interatomic potentials) and/or the AFIR method.
- Solid background in DFT and reaction-mechanism studies; experience with transition-metal, lanthanide or actinide chemistry is an asset.
- Proficiency in Python and scientific ML (PyTorch, generative models, chemical descriptors) is valued.
- Autonomy, collaborative mindset and good scientific communication in English.

Environment

The successful candidate will join a technically proficient team with in-house expertise in ANN, VAE and DFT surrogate models, and will benefit from an allocation of approximately 2 million CPU/GPU hours per year at the [CALMIP regional supercomputing centre](#) (recently renewed HPC infrastructure).

Salary

Gross salary is commensurate with experience, according to CNRS / university scales.

Indicative net monthly salary: $\approx [2500 - 2800]$ € net/month.

Application

Applications should include a CV, a publication list, a brief statement of research interests, and the contact details of two referees. Please send them to:

Prof. Laurent Maron, laurent.maron@utoulouse.fr

Prof. Romuald Poteau, romuald.poteau@utoulouse.fr

Applications will be reviewed on a rolling basis until the position is filled.

Non-exhaustive bibliography

- (a) L. Escomel, I. del Rosal, L. Maron, E. Jeanneau, L. Veyre, C. Thieuleux & C. Camp (2021), Strongly Polarized Iridiumupdelta—Aluminumupdeltamathplus Pairs: Unconventional Reactivity Patterns Including CO₂ Cooperative Reductive Cleavage, *J. Am. Chem. Soc.* **143**: 4844-4856, [10.1021/jacs.1c01725](#); (b) X. Shi, T. Rajeshkumar, L. Maron & J. Cheng (2022), CO, CO₂ and CS₂ activation by divalent ytterbium hydrido complexes, *Chem. Comm.* **58**: 1362-1365, [10.1039/d1cc06449e](#); (c) T. Simler, K. N. McCabe, L. Maron & G. Nocton (2022), CO reductive oligomerization by a divalent thulium complex and CO₂-induced functionalization; *Chem. Sci.* **13**: 7449-7461; (d) K. Kunihiro, T. Rajeshkumar, L. Maron, S. Heyte, S. Paul, T. Roisnel, J.-F. Carpentier & E. Kirillov (2025), On the Mechanism of Acrylate and Propionate Silyl Esters Synthesis by Ruthenium-Catalyzed Coupling of CO₂ with C₂H₄ in the Presence of Hydrosilanes: Combined Experimental and Computational Investigations, *Mol. Catal.* **570**: 114660, [10.1016/j.mcat.2024.114660](#)
- a) O. Santoro, L. Piola, K. M. Cabe, O. Lhost, K. Den Dauw, A. Vantomme, A. Welle, L. Maron, J.-F. Carpentier & E. Kirillov (2020), Long-Chain Branched Polyethylene via Coordinative Tandem Insertion and Chain-Transfer Polymerization Using rac-EBTHiZrCl₂/MAO/Al-alkenyl Combinations: An Experimental and Theoretical Study, *Macromolecules* **53**: 8847-8857; (b) Y. Zhong, I. Douair, T. Wang, C. Wu, L. Maron & D. Cui (2020), Access to Hydroxy-Functionalized Polypropylene through Coordination Polymerization, *Angew. Chem. - Int. Ed.* **59**: 4947-4952, [10.1002/anie.201914747](#); (c) O. Santoro, L. Piola, K. Mc Cabe, O. Lhost, K. Den Dauw, A. Vantomme, A. Welle, L. Maron, J.-F. Carpentier & E. Kirillov (2021), Al-alkenyl-induced formation of long-chain branched polyethylene via coordinative tandem insertion and chain-transfer polymerization using (nBuCp)2ZrCl₂/MAO systems: An experimental and theoretical study, *Eur. Polym. J.* **154**: 110567, [10.1016/j.eurpolymj.2021.110567](#); (d) Z.-H. Zhang, Y. Sun, T. Rajeshkumar, Y. Li, L. Maron & M. Hong (2025), Vinyl polymers with fully degradable carbon backbones enabled by aromatization-driven C-C bond cleavage, *Nat. Chem.* **17**: 746-755, [10.1038/s41557-025-01751-w](#)
- (a) T. Fages, F. Jolibois & R. Poteau (2021), Recognition of the three-dimensional structure of small metal nanoparticles by a supervised artificial neural network, *Theo. Chem. Acc.* **140**: 98, [10.1007/s00214-021-02795-0](#); (b) S. Christodoulou, C. Cousseau, E. Limanton, L. Toucouere, F. Gauffre, B. Legouin, L. Maron, L. Paquin & R. Poteau (2024), Efficient Machine-Learning-Based New Tools to Design Eutectic Mixtures and Predict Their Viscosity, *ACS Sustainable Chem. Eng.* **12**: 18537-18554, [10.1021/acssuschemeng.4c05869](#); (c) S. Christodoulou, E. Dos Santos, R. Poteau & L. Maron (2025), Styrene Polymerization with Group III Metallocenes: A Generative and Interpretable Catalyst Design Strategy Combining DFT and Machine Learning, *ChemistryEurope* e202500250, [10.1002/ceur.202500250](#)
- See H. J. Kulik & M. S. Sigman (2021), Advancing Discovery in Chemistry with Artificial Intelligence: From Reaction Outcomes to New Materials and Catalysts, *Acc. Chem. Res.* **54**: 2335-2336, [10.1021/acs.accounts.1c00232](#) and references therein (special issue "Data Science Meets Chemistry")